

Reducing Cognitive Translation Cost in Embodied Tactile Robot Control with a Learned Spatiotemporal Policy

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Abstract. When an operator guides a collaborative robot, the desired motion is perceived directly at the robot body, yet conventional interfaces force this spatial intention through a chain of symbolic translations—mode selection, key lookup, coordinate-frame reasoning—before any motion occurs. We frame this burden as *cognitive translation cost* and argue that an embodied tactile interface, where the operator slides a finger over a capacitive skin on the robot itself, shortens the intention-to-action pathway. That pathway, however, is only as strong as its machine-decoding stage, which must remain trustworthy on a physical sensor that drifts, ages, and breaks. The core contribution of this paper is a learned decoder for this stage: from operator demonstration sessions recorded directly on the platform, with no manual labeling, we train a compact spatiotemporal policy that maps raw capacitance-map sequences to velocity commands and interaction modes, using geometric augmentation that exploits the sensor's physical symmetries and perturbation-aware augmentation that corrupts the inputs while keeping the command targets fixed. On a faithful-replay robustness benchmark that streams identically perturbed episodes through unmodified deployed implementations, the learned policy reproduces the demonstrated commands on clean input and outperforms a conventional hand-crafted centroid-tracking controller in every tested condition across four perturbation families. A planned user study will quantify the remaining human-side translation components against keyboard and teach-pendant control.

Keywords: Human-robot interaction · Embodied interaction · Tactile sensing · Imitation learning · Robustness evaluation · Cognitive translation cost

1 Introduction

Collaborative robots are increasingly operated by people who stand next to them. The operator sees, at the robot or the workpiece, exactly which way the

tool should move—yet most interfaces require that spatial intention to be re-expressed symbolically: find the correct key or pendant button, verify the coordinate frame, check the control mode, then watch whether the response matches the intent. We use the term *cognitive translation cost* for the observable burden accumulated along this intention-to-action pathway, measured through delays, wrong-axis selections, mode transitions, corrective inputs, and response latency (Sect. 3).

An *embodied* tactile interface attacks the translation problem at its source: a capacitive skin wrapped around a robot link lets the operator place a finger on the robot and slide it in the direction the tool should move, so the command action is physically co-located with the robot and spatially congruent with the intended motion [3]. But co-location only removes the *human-side* translation stages. The pathway still contains a machine-decoding stage that converts raw taxel data into velocity commands, and conventional implementations are hand-crafted pipelines of centroid tracking, thresholds, and mode heuristics whose every parameter silently assumes a clean, calibrated, undamaged sensor—while capacitive arrays drift, taxels die, and gain varies across fabrication batches. An embodied interface whose decoder misreads a degraded sensor does not reduce translation cost; it merely relocates it into corrections and re-tries.

This paper therefore concentrates on the machine side of the pathway and makes it *learned, compact, and measurably robust*. Our contributions are:

1. **A conceptual framing** that decomposes the intention-to-action pathway into four observable translation components and locates the machine-decoding stage within it (Sect. 3).
2. **An automatic demonstration-recording pipeline**: labeled training episodes are logged as a by-product of normal use, with no manual labels (Sect. 4).
3. **A location-preserving learned policy**: a convolutional encoder with per-channel spatial softmax and a recurrent head, trained with geometry-aware and perturbation-aware augmentation that corrupts the inputs while keeping the command targets fixed (Sect. 5).
4. **A faithful-replay robustness benchmark** that scores unmodified deployed controller implementations under identical sensor corruptions, showing the learned policy outperforms a conventional hand-crafted centroid-tracking controller in every tested condition (Sects. 6–7).

A user study comparing the embodied interface against keyboard and teach-pendant control is specified in Sect. 8.

2 Related Work

Human–robot interaction research has long examined how interfaces bridge the gap between human intention and robot representation [1, 2]: teach pendants provide structured industrial control but require coordinate-frame interpretation, while kinesthetic teaching reduces representational distance but depends on backdrivability or force sensing. Spatial-compatibility research supplies the

underlying mechanism—responses are faster and less error-prone when stimulus and response arrangements are spatially congruent [3]. Robot tactile sensing has meanwhile progressed from discrete contact switches to conformal and textile skins [4–6]. Most systems treat touch as information *received* by the robot; a smaller body of work treats it as an intentional command channel [7], typically realized with hand-crafted centroid-tracking controllers for continuous velocity guidance. We are not aware of prior work that both learns this mapping end-to-end from the raw taxel stream and quantifies its robustness against the conventional controller design under identical sensor corruptions.

Casting controller construction as supervised learning from recorded demonstrations is a long-standing idea [8, 9], with well-understood strengths and caveats such as covariate shift [10]. Here the demonstrations are recorded on the deployed platform itself, so the training distribution matches deployment by construction; the open question is generalization under *input corruption*, which is exactly what our benchmark measures. Spatial-softmax feature points originate in end-to-end visuomotor learning [11, 12]; we show they matter even on a coarse taxel grid, where global pooling destroys the location information that determines command direction.

Finally, aggregate outcomes such as completion time establish whether an interface works, but compress several sources of inefficiency into one number. Cognitive-infocommunications research argues for analyzing human–machine systems through the structured interplay of their cognitive capabilities rather than end-to-end scores alone [14], and corruption benchmarks report robustness per perturbation family rather than as a single score [13]. We apply this principle to both the human side (observable translation stages) and the machine side (robustness per perturbation family and magnitude).

3 Cognitive Translation-Cost Framework

Let I denote the user’s intended robot motion, D the interface-compatible command decision, U the physical input action, C the machine-decoded command, A the resulting robot action, and F the observable feedback:

$$I \rightarrow D \rightarrow U \rightarrow C \rightarrow A \rightarrow F. \quad (1)$$

Translation burden does not arise at a single point; different interfaces load different stages (Fig. 1). Four observable components summarize the pathway: *mode-switching cost* M (interface-state transitions before a command can be issued), *command-production cost* N (physical input events including corrections), *spatial-translation cost* S (difficulty of mapping a perceived task-space motion to the interface representation), and *feedback-closure cost* L (delay from input to observable robot response).

For keyboard and pendant interfaces the decoding stage is trivial—a key press is unambiguous—and translation cost concentrates in S , M , and N . An embodied tactile interface inverts this distribution: touching the robot where motion should occur makes S nearly free and collapses mode navigation, but

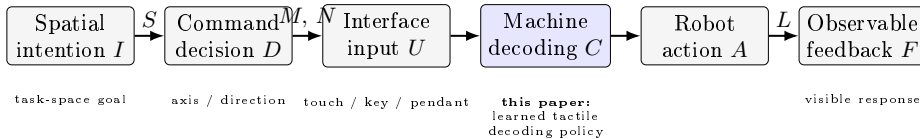


Fig. 1. Intention-to-action pathway. Embodied tactile input targets the human-side costs S , M , and N ; this paper concentrates on the highlighted machine-decoding stage $U \rightarrow C$, whose brittleness under sensor degradation would re-enter the pathway as corrective inputs.

decoding becomes a perception problem on a physical sensor that drifts, ages, and breaks. Reliability of stage C is thus the condition under which the embodied interface’s human-side advantage survives contact with reality; it motivates the learned decoder and the explicit robustness benchmark that follow.

4 Embodied Tactile Platform and Demonstration Data

The testbed is a woven capacitive tactile skin providing an $n \times m$ grid of taxels, mounted on the end link of a 6-DOF collaborative arm (TM5-900); interlaced conductive threads conform to the curved robot geometry. Per frame the skin outputs a matrix of relative capacitance changes against a per-session calibration baseline; touch produces negative deviations roughly proportional to contact pressure. A short moving average is computed alongside the instantaneous signal, and frames are consumed at approximately 60 Hz. The command vocabulary covers task-space translation driven by finger motion, plus two pressure-gated modes: a hard press commands a *push* along $+v_y$ and a two-finger pinch commands a *pull* along $-v_y$. **[One to two sentences on skin construction and readout electronics, with a citation to the hardware publication if available.]**

During demonstration sessions an operator guides the end-effector by interacting freely with the skin, and at each control tick the platform automatically records the instantaneous and averaged capacitance matrices, the commanded 6-DOF end-effector velocity, the interaction mode $m_t^* \in \{\textit{stop}, \textit{move}, \textit{push}, \textit{pull}\}$, and lightweight contact descriptors computed from the touch map (centroid, displacement, speed, peak and mean pressure, touch presence). Labeled training episodes are thus a by-product of normal operation; no frame was manually annotated. The corpus comprises eight episodes—roughly five minutes of continuous interaction—across a free-swiping session and a push/pull-focused session. Rotational axes were not exercised and are locked to zero throughout.

5 Learned Decoding Policy

5.1 Input Representation and Architecture

The policy consumes a sliding window of recent frames, each a 4-channel $n \times m$ image: the instantaneous capacitance deviation, its moving average, the temporal difference between consecutive frames, and a binary touch mask obtained by thresholding the averaged channel. A compact auxiliary vector carries the contact descriptors of Sect. 4, which summarize contact geometry while the raw frames preserve everything they discard.

A stack of 3×3 convolutions processes each frame at full taxel resolution—no spatial pooling at any depth. A per-channel *spatial softmax* then reduces each feature map $F_k \in \mathbb{R}^{n \times m}$ to its expected image coordinate,

$$\mathbf{p}_k = \sum_{(i,j)} \text{softmax}(F_k/\theta)_{ij} (u_i, u_j), \quad u \in [-1, 1], \quad (2)$$

with a learned temperature θ . Concatenating the keypoints with the per-channel mean intensities yields a frame descriptor that preserves *where* the contact is and *how hard* it presses; the common alternative, global average pooling, collapses location entirely, and Sect. 7.3 quantifies its cost. Frame descriptors are fused with an encoding of the auxiliary vector and passed through a single-layer GRU, whose final hidden state feeds a velocity-regression head and a mode classifier. The model is compact and its per-window CPU inference time sits an order of magnitude inside the 60 Hz control budget, contributing negligibly to the feedback-closure cost L .

5.2 Geometry- and Perturbation-Aware Augmentation

Five minutes of demonstration cannot cover the product space of contact locations, directions, and pressures—but the sensor’s geometry provides exact symmetries. During training each sampled window is randomly *flipped* on the taxel grid, with the transformation propagated consistently through every affected quantity (a column flip negates v_x and mirrors the column-related contact descriptors; a row flip acts analogously on v_z), and *translated* by a small random offset bounded so that no active cell leaves the grid. Flips multiply directional coverage by four; translation decouples the mapping from absolute contact position; mode labels are invariant throughout.

A second, *perturbation-aware* scheme targets sensor degradation directly. For a random subset of training windows, one of two corruption modes is sampled, mirroring the one-failure-at-a-time structure of the benchmark: either i.i.d. Gaussian noise of a randomly drawn magnitude, injected exactly as physical noise would propagate through the sensing pipeline (raw noise into the instantaneous channel, its moving average into the averaged channel, its delta into the temporal-difference channel, and the touch mask recomputed from the corrupted average), or a randomly drawn baseline offset that emulates calibration

drift. Applying at most one corruption family per window proved important: stacking noise and drift compounds the ambiguity and makes the policy hedge its velocity magnitudes even on clean input. Crucially, the velocity and mode *targets are left unchanged*—the operator’s finger is doing the same thing regardless of sensor health—so the policy learns to decode the same intent from a degraded stream; Sect. 7.3 shows this is what closes the noise-robustness gap. Augmentation is never applied to validation data.

5.3 Training Protocol

Windows are split by *episode*: six episodes train, two held-out episodes (one per session) validate, and normalization statistics use training data only. The loss combines a Smooth- L_1 velocity term with a class-balanced cross-entropy mode term, weighted per sample to de-emphasize *stop* frames and emphasize the rare *pull* class; training uses AdamW with plateau-based decay and early stopping.

6 Faithful-Replay Robustness Benchmark

The benchmark answers one question: *when the sensor misbehaves, which decoder keeps translating operator intent correctly?* The learned policy is compared against a conventional **hand-crafted centroid-tracking controller**, representative of how continuous tactile guidance is normally implemented: contacted taxels are clustered, a pressure-weighted centroid is tracked with hysteresis and deadbanding into velocity, and pressure-gated heuristics arbitrate the push/pull modes—dozens of tuned parameters, each silently assuming a healthy sensor. Three design decisions distinguish the benchmark.

(1) *Deployed implementations, not re-implementations.* Recorded episodes are streamed frame-by-frame into a sensor-emulation layer, and both controllers run through their unmodified runtime classes, with all stateful machinery, buffering, normalization, and confidence gating.

(2) *Ground truth from clean input.* The reference at frame t is the command recorded on the *clean* sensor stream, while both controllers receive an identically *perturbed* stream; a controller scores well if corruption does not change what it believes the operator wants.

(3) *Self-validation.* At zero perturbation the replay harness reproduces the commands recorded live on the held-out episodes, closing the loop on harness fidelity before any perturbation is trusted.

Four perturbation families model the dominant failure modes of capacitive arrays, injected consistently into both the instantaneous and moving-average channels at mild and severe operating levels: **baseline drift**, a uniform offset that emulates thermal/humidity drift between calibrations (beyond the severe level, idle cells begin to register as touched and sensing fails for any decoder); **dead taxels**, a random fraction of cells clamped to zero (up to 20%); **gain error**, multiplicative scaling (0.5–1.3 $\times$); and **Gaussian noise**, i.i.d. per-cell interference. Velocity error is reported as linear-axis RMSE against the clean-input

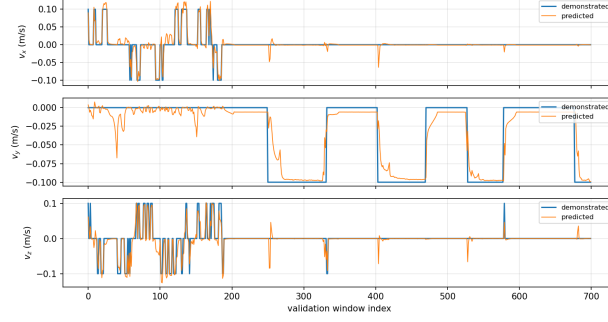


Fig. 2. Predicted vs. demonstrated velocity on a held-out validation segment (v_x , v_y , v_z from top to bottom). The policy reproduces direction changes on all three axes, push/pull events (v_y), and stop phases.

reference command; mode agreement as the fraction of frames whose canonical mode matches.

7 Results

7.1 Fidelity to the Demonstrated Commands

On the two held-out episodes the policy attains a linear-velocity RMSE of **1.55 cm/s** against demonstrated commands whose magnitudes reach 10 cm/s, and **92.6%** mode accuracy; mode confusions concentrate where the demonstrations themselves are ambiguous—*stop* frames bordering *push* onsets (Fig. 2). The policy thus faithfully reproduces the demonstrated behavior on clean input; whether it holds on a degraded sensor is answered next.

7.2 Robustness under Sensor Perturbation

Table 1 and Fig. 3 report the benchmark.

The learned policy wins every tested condition. Across all four perturbation families and all magnitudes, the policy’s velocity error stays well below the centroid controller’s, and its mode agreement stays above it throughout. Under baseline drift the policy’s error barely rises; with one in five taxels dead it is nearly unchanged—the convolutional encoder interpolates around missing cells, whereas a pressure-weighted centroid has no such mechanism. The centroid controller’s modest mode agreement even on clean input is not a replay error: the recorded reference captures the operator’s *intended* interaction, which the heuristics do not always match.

Noise robustness had to be trained, not assumed. Uncorrelated per-frame noise is structurally hostile to this input representation: the temporal-difference channel doubles the variance of i.i.d. noise and the binary touch mask flickers under borderline contact readings, while the centroid controller is partially shielded

Table 1. Faithful-replay benchmark on held-out episodes: velocity RMSE (m/s, lower is better) and mode agreement (higher is better) against the clean-input reference. Bold marks the better controller per condition.

Perturbation	Level	RMSE (m/s)		Mode agree.	
		Centroid	Learned	Centroid	Learned
None (clean)	–	0.032	0.015	0.68	0.93
Baseline drift	mild	0.032	0.017	0.69	0.91
	severe	0.033	0.018	0.71	0.91
Dead taxels	10%	0.032	0.015	0.68	0.93
	20%	0.034	0.018	0.66	0.91
Gain error	0.7×	0.036	0.019	0.61	0.90
	1.3×	0.029	0.016	0.75	0.92
Gaussian noise	mild	0.034	0.018	0.66	0.90
	severe	0.040	0.031	0.66	0.75

Table 2. Ablation on robustness (velocity RMSE, m/s; lower is better); each row adds one ingredient. The hand-crafted centroid controller scores 0.033–0.034 under these noise conditions. **[Re-run the first row as a controlled single-variable ablation before submission.]**

Variant	Clean	Mild noise	Stronger noise	Dead 20%
GAP encoder, no geom. aug.	0.014	0.055	0.059	0.022
+ spatial softmax, geom. aug.	0.014	0.033	0.043	0.019
+ perturbation-aware aug. (ours)	0.015	0.016	0.018	0.018

by reading only the moving average. A variant trained *without* perturbation-aware augmentation indeed falls behind the centroid controller under noise (Sect. 7.3); injecting sensor-consistent noise and drift during training closes this gap completely, at almost no cost on clean input.

7.3 Ablation: Location-Preserving Encoding and Augmentation

Table 2 isolates each design choice. Clean-input fidelity is essentially identical across all three variants—an evaluation reporting only clean held-out metrics would find nothing to choose between them. Under perturbation the differences are large: the location-preserving encoder and geometric augmentation cut the noise-induced error by roughly 40% relative to global average pooling but still trail the hand-crafted controller under noise, and perturbation-aware augmentation closes the remaining gap and reverses it at a negligible cost in clean RMSE. Robustness, not clean accuracy, is where these design choices pay off.

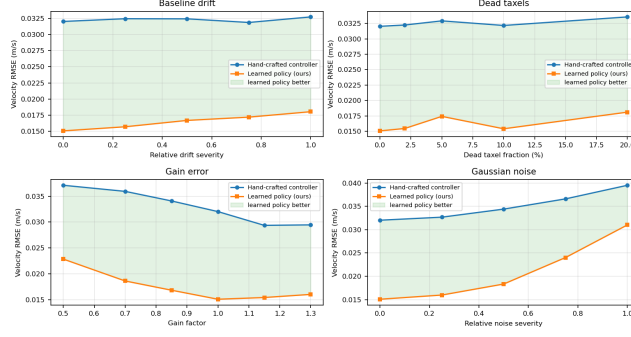


Fig. 3. Velocity RMSE vs. perturbation magnitude for both controllers. Green shading marks conditions where the learned policy has the lower error: every tested condition of all four families.

8 Planned User Evaluation

The results above establish that the machine-decoding stage is robust enough to carry a user-facing comparison. A ready-to-run protocol will quantify the human-side components: three matched interfaces—embodied tactile control using the learned policy, a keyboard, and the teach pendant—issue the same bounded velocity primitives, with randomized spatial command translation as the primary task and short multi-command sequences as a secondary task. The principal measure is intention-to-command initiation latency, complemented by first-input correctness and wrong-axis selections (S), interface-state transitions (M), corrective inputs (N), and input-to-motion delay (L), reported as a profile that preserves *where* each interface loads the pathway. A sub-condition with a masked-off taxel region will test whether decoder degradation manifests as increased correction burden. **[Effect sizes, participant count, and statistics after data collection.]**

9 Limitations and Conclusion

Learning from demonstration reproduces the demonstrated strategy rather than discovering better ones; our claim is robustness under corruption, not superiority of the strategy itself. The augmentation uses i.i.d. noise and uniform drift; correlated noise models fitted to the physical sensor are a natural refinement. All robustness results are open-loop replay, a gap the user study closes. Rotational axes were locked to zero, and results come from a single sensor unit and few operators.

In sum, an embodied tactile interface shortens the human side of the intention-to-action pathway, but its promise rests on the machine-decoding stage remaining trustworthy when the sensor does not. We showed that this stage can be learned directly from operator demonstrations recorded on the platform,

and that the resulting policy outperforms a conventional hand-crafted centroid-tracking controller in every tested condition across all four perturbation families. The methodological lesson from the ablation: this robustness was invisible to clean held-out evaluation and had to be created deliberately, by corrupting training inputs while holding the command targets fixed. A decoder that holds its ground under sensor degradation is what keeps the embodied interface’s cognitive advantage intact.

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